

Opinion of the scholars regarding the scientific miracles of al-Quran.

Several scholars have expressed their points of views regarding this topic *Jawaheri* is one of them. He asks why the scholars have compiled thousands of books on Fiqh, whereas there are no more than 150 verses related to the Islamic Legal system as revealed in al- Quran. On the other hand there are more than 750 verse related to natural science. These are obviously which are related directly to these phenomenon, what about the rest which implies the scientific facts rather than speaking direct. So doesn't it look harsh that this nation remains involved in limited number of verses ignoring the majority?

Our forefathers mastered the legal studies which was prevalent at that time. But if they were alive today they would have given their lives for the study of modern science so that the nation may witness a faster pace of prosperity.

Opinion of Dr. Muhammad Abdullah Daraz:

Addressing this issue he wrote a book entitled "Al Bina al-Azim" in the year 1933 A.D. Here he added that al- Quran contains three kinds of miracles:

One that is related to literature. The second one is related to science and the third one is related to law.

Meaning of scientific miracles

All the messengers are sent with miracles suitable for his nation, time and message. The prophets before Muhammad (SM.) used to emerge to a particular nation for a particular nation for a particular period. So Allah has sent them with miracles that were mostly material and ephemeral. Such as, the stick of Musa (A.M.) and making the dead living again by the touch of Isa(A.M.) When the

power of these miracles diminished by the tide of time and people got astray by distorting the original message Allah used to send another messenger with new miracles fit for his age but ephemeral and short living.

But the case is quite different for the last Prophet Muhammad (S.M.) Allah decided to bestow him with messages that should be comprehensive and immortal in order to produce ever renewing guidance fit for all the times all the nations to come.

In the beginning of the 19th century science has become very much instrumental and more experimental. Scientists began to explore the secrets of the universe in and outside of the world. The more they revealed the secrets of the world the more they confirmed the truth of al-Quran which was revealed fourteen hundred years ago. To our great surprise this fact was confirmed by al-Quran at the very beginning:

"We will show them Our signs in horizons and within themselves until it becomes clear to them that is the truth" (41/53)

The signs of Allah in this verse have three meanings:

First: The creations created by Allah in the sky and the earth as Allah said:

"And of his assigns in the creation of the heavens and the earth and what he has dispersed through out them of them of creatures."(42/29)

Second: the verses of al-Quran that describes the variations of the creation and they are so many.

Third: The evidence and miracles sent by Allah to support his prophet (S.M.) in various occasions and places reflected in the secrets of the nature intermittently,

Explaining the verse Qurtubi said in his Tafsir:

Ata said the meaning of "Horizons" in the verse that they are signs of Allah in the heaven such as the sun, the moon, the stars, the day and the night, the wind and all the things on the earth such as the cities, plants, mountains, thunders and oceans e.c.t"

Explaining the later part of the verse Jalalain said:

And the signs in themselves indicate the secrets of the harmonious creation.

So these are the signs described in the book of Allah. Their meanings are realized only now, centuries after the revelation by virtue of the scientific findings to prove that al-Quran is the truth revealed by Allah.

Some examples of the verses related to scientific miracles:

1) Extension of the Universe;

And the heaven We constructed with strength, and indeed We are its expanders.”(51/47)

Einstein has said the universe is extending with all its glittering stars. Natural scientists have discovered that the universal substances are separating from each other and the distance is increasing day by day.

So we can realize that the universe is not a substance that lies fixed but indeed it is extending day by day to the contrary of many material contents which never grow more than a certain limit.

In this regard Dr.Hubble said:

There is one common feature of the scattered stars. They move faster backward than forward, and the speed increases with the increase of the distance from each other.

Division of the atom:

It said in al-Quran:

“And not absent from your lord is any part of an atoms weight within the earth or the heaven or anything smaller than that or greater than that or greater but that is in a clear register.”(10/61)

Until the 19th century it was believed that the atom could not be divided any way. But they tried hard to find some theory contrary to that. At last only few decades ago they were able to

Estimate of Al-Mamun : Al-Mamun, the most distinguished figure of the Abbasids, Civilization and culture, State Council of Mamun:

Al-Mamun was unquestionably the most distinguished figure of the House of Abbas, nay of the whole Islamic World. He was not only a brave soldier but also a good administrator. He is will to work, his prudence, his clemency and judgment, his sagacity and liberalit3T the chief features of his character. His reign forms the most glorious epoch in the history of Islam. "The twenty years of his reign have left enduring monuments of the intellectual development of the Muslims in all directions of thought." The intellectual advancement was not restrictions to a particular branches but it pervaded the whole culture domain. Mathematics, astronomy, medicine and philosophy made and extraordinary progress during this period. He fully realized the truth that the actual happiness of his people lay in education and culture and he translated his realization into practice by opening schools and colleges in all directions. He made no distinction of creed or race and threw public offices open to all. Liberty of conscience and religion was given to all. He lessened the religious autocracy by establishing a Council of State of advice the Caliph. It was composed of representatives from all the communities under his rule, including Muslims, Jews, Christians and others.

The Mu'tazlites:

Mamun interested himself in question of religious dogma. He had embraced the Mu'tazlite doctrines about free will and predestination and was in particular shocked at the opinion which had spread among the Muslim doctors that the Quran was the uncreated word of God. In the year 212 A.H./827 A.D., he published an edict by which the Mu'tazlite doctrine was declared to be the religion of the state and the orthodox faith was condemned as heretical. At the same time he ordered all his subjects to honour Ali as the best creature of God after the Prophet and forbade the praise of MuawiJn 218 A.H/833 A.D. a new edict appeared by which all judges and doctors were summoned to renounce the error of the uncreated word God. Most of them expressed their agreement with the views of the Caliph, but a few stood firm in there conviction among whom was Ahmad bin Hanbal(founder of the Hanbali School),who was ordered to be put into prison. Two of Mamun's successor's maintained the edict and Ahmad bin Banbal was flogged ultimately at their hands.

Science and literature, Assembly of learned men, Works of translation, Observatory established:

Mamun's reign was the most glorious and brilliant of all in the intellectual history of Islam. His reign was the revival of learning both in the East and in the West. At this Court were munificently entertained men of science and letters, poets, physicians and philosophers. Besides being an age of the philologist and a collectors of traditions, such as the great Bukhari and of historian as al-Waqidi". Moreover, the 17the Christians were welcomed at the court not only for their learning but for being well versed both in Arabic and the language and literature of Greece. He completed the works of translation started

9.2

by his grandfather Mansur. Under his guidance Sanskrit books, Greek Mathematics and philosophy, science of Euclid and Ptolemy were translated into Arabic. Costa, son of Luke, was appointed for the translation of Greek and Syrian; Yahya bin Harun, of Persian; and Duban the Brahman, of Sanskrit works into Arabic. "It was through the labours of these learned men", says W. Muir, "that the nations of Europe, then shrouded in the darkness of the Middle Ages, became again acquainted with their own proper but forgotten patrimony of Grecian science and philosophy", the Persian language received a great impetus in his hand. An observatory was established in the plain of Tadmör for the study of Astronomy and geometry. Astronomical observations made great stride during this reign. Abul Hasan invented the telescope from a tube. Poet Abbas, founder of the modern Persian poetry, and al-Kindi, employed at the famous 'Bayt al-Hikmah established by Mamun in Baghdad, and translator of Greek works, flourished during this period.

Philosophical re-union:

Mamun was so interested in philosophical discussions that he sat apart Tuesday fully for the same purpose. The servants and scholars used to attend his chamber regularly and the Caliph satisfied them with his philosophical conversation. Thus taking all these points into consideration, the historians are of opinion that the reign of Mamun constitutes the most glorious epoch of Saracenic history and has been called the 'Augustan Age of Islam.'

Intellectual awakening of the Muslims:

With the expansion of Islamic territories, the Muslims were brought to the cultural heritage of the foreign countries. It was the influence of the Persio-Hellenic culture that helped the Muslims to play an important role in the intellectual firmament of the world. In this connection P.K. Hitti says, "the awakening was due in large measure to foreign influence, partly Indo-Persian and Syrian, but mainly Hellenic and was marked by translation into Arabic from Persian, Sanskrit, Syriac and Greek".

Era of translation: Age of translation followed by original contributions:

The Muslims had translated every branch of knowledge either from Greek or from Sanskrit or from Pahlavi (Middle Persian) before they had assimilated it, before the establishment of the famous Bayt al Hikmah as became the main centre of translation work as most of the translators were Aramaic speaking, many of the works during this period were translated into Aramaic (Syriac). But it was not continued for a long time. The Greek work soon began to be translated into Arabic. Harun ar-Rashid appointed learned translators of all nationalities and creeds-Hindus, Parsis, Christians, Jews and Muslims. Books and extant materials were collected by Mamun from all countries. He is said to have asked the Byzantine Emperor to send the savant Leo to Baghdad in exchange for five tons of Gold. The Muslims became acquainted with the philosophical lore and the medical literature of Greece and the scientific works of Persia and India. They translated many foreign works into Arabic and thereby preserved the culture of the ancient world. "The researches of Aristotle, Galen and Ptolemy would have been lost to the world, if the

Muslims had not kept them preserved by translation". The age of translation was soon succeeded by a period of original thinking and research. The Muslims sent their translation together with their own contributions to Europe. Modern Europe owes much to them in Chemistry, medicine and mathematics.

Architecture:

With the fall of the Umayyad dynasty, the seat of the Caliphate was transferred from Damascus to Baghdad and the Persian. Influence began to have a marked effect on the manners and habits of the Muslim World. The Arabs lost their pre-eminence not only in the Army and at court, but also in the society. Their cultural life was also affected. The Hellenistic influences of Syria were replaced by the influences of Sasanian Persia and Iraq with profoundly modified the art and architecture.

Variation of Design:

The Abbasid Caliphs, like the Umayyad patronized art and architecture. During the reign of more than five hundred years many buildings and places were built in different parts of the Empire. The second Abbasid Caliph, al-Mansur, founded the city of Baghdad in the year, 762 A.D. The city was noted for a good number of places, mosque and other building built by successive Caliphs and other reigning princes of Iran. It was round-shape city enclosed by a double wall of four gates, namely the Kufa Gate (S.W) the Basrah gate (S.E), the Khurasan Gate (N.E) and the Damascus Gate (N.W.). The Caliph shifted his residence to Baghdad when it was made the capital of the Abbasid empire. The main features of the newly founded city were the palaces of the Caliph, called the Golden Gates, crowned by a great dome of green colour. The palace was a square of 400 cubits and side and lay in the middle of the city. A second place was called the Khuld, was constructed after a few years and this became a favourite residence of Harun ar Rashid. Al — Mansur built the royal mosque. It was built of sun fired bricks and clay. Harun-ar-Rashid rebuilt it and Mutazid enlarged it. The mosque of Cordova (787 A.D.) The great mosque of 'Amr in Egypt (827 A.D.) The mosque of Cairowan. (836 A.D.), the great mosque of Susa (850 A.D.) The great mosque of Samara and mosque of Turns speak much of the architectural taste of the Abbasids.

None of the architectural monuments left to day:

Azad-ud-Dawlah, the greatest Buwayhid Ameer, built a fine hospital on the bank of the Tigris. It was more a palace than a hospital. A new palace called the Taj was constructed by the Caliph Mutazid. It was completed by his successors. There were many other magnificent palaces and buildings and many wealthy people, princes and even Emperors of Iran had their own palaces in Baghdad. But unfortunately no trace of the architectural monument which once adorned the city of Baghdad, is left to the present day. In this connection Prof. Hitti says, "The Caliphal palace called the Golden Gate or Green Dome created by the founder of Baghdad, as well as his palace of Eternity al-Khuld) and the Rusafah palace, built for the crown prince alMahdi, the palace of the Plediades on Samara, spent 40,00,000 dinars. his adjoining palace styled the Crown (al-Taj), completed by his son al-Muqtadir, designated the Hall of Tree on account of the gold

and silver tree that stood in its pond; the Buwayhid Mansion known by the name al-Muizziyah after Muizz ud-Dawlah which cost 10,00,000 dinars all these and others like them left no remains to give us an inkling of the splendour that was theirs, "The destruction was brought about by the civil war between Amin and Mamun, by the invasion of Halagu and by other natural causes.

Scientific and literary activities during the Abbasid period:

The Abbasid Caliphs were great patrons of learning and the learned. They nourished and nurtured a galaxy of brilliant scientist and scholars who made their most valuable contributions to the culture of the world. the following subjects will speak of their contributions to different branches of science.

Science and Medicine:

The Umayyad Caliphs encouraged the study of medicine but real progress in Arabian medicine began with the coming of the Abbasids. The early Abbasids Caliphs established hospitals called "bimaristan". The first was opened by Harun arRashid in Baghdad. In course of time "thirty four hospitals were established in various parts of the Muslim world", other branches of medicine, such as surgery, pharmacy, ophthalmology, etc. were greatly developed during the Abbasid period. The notable medical authors of this period were Persian in nationality but Arab in language. Au al-Tabari, al-Razi, Ali ibn Abbas and Ibn Sina occupy unique places in the history of the world.

Al- Razi, great physician of the Islamic world:

The next outstanding figure in the medical field was Abu Bakr Muhammad ibn- Zakaria al-Razi (European called Rhazes) who was born at Rayy near Tehran. AlRazi was unquestionably the greatest physician during the Middle Ages and one of the greatest physicians of all time. He was a student of Hunayn ibn Ishaq who was well acquainted with Greek, Persian and Indian medicine. Early in life, he learnt music and could play well on the lute. in his youth he practiced as an alchemist but in his later years when his reputation attracted pupils and patients from all parts of western Asia, he devoted himself exclusively to medicine. He wrote as many works as two hundred, half of which are on medicine. One of the most celebrated works of al- Razi is his treatise, 'on Smallpox Measles' (al-Judari al-Hadhab) which was early translated into Latin and later into various languages including English. It was printed not less than forty times between 1498 and 1866. This treatise was considered an ornament to the medical literature of the Arabs. It served to establish al-Razi's reputation as one of the keenest original thinkers and greatest clinicians not only of Islam but of the Middle Ages. But greatest of his medical works is al-Hawi (the Comprehensive Book) which was written in twenty volumes. For each disease, he first cites all the Greek, Syrian, Arabic, Persian and Indian authors and at the end he gives his own opinion and experiences. This work was translated into Latin under the auspices of Charles I of Anjou by the Sicilian Jewish physician Faraj Ibn Salim in 1279 A.D. under the title of Continens. It was printed several times from 1484 onwards. Its influence of European medicine was very considerable.

Al-Razi, was also mathematician:

Al-Razi was the chief physicians at Bagdadh hospital. He is also considered the inventor of the section in surgery. Besides medicine, al-Razi left writings on theology, mathematics, natural science and astronomy. The last but one deals with matter, space, time, motion, nutrition, growth, putrefaction, meteorology, optics and alchemy. One of the principal works on alchemy was the kitab al-Asrar (The Book of Secrets) which was translated into Latin by the distinguished translator, Gerard of Cremona. Al Razi had a prominent contemporary known to the west as Isaac Judaeus who was the Fatimid Caliph of Qairouan in Tunisia.

All ibn al-Abbas:

'Au ibn Abbas (Hayy Abbas) was a Persian Muslim of Zoroastrian descent. He wrote an encyclopedia named "the whole medical Art", known to the Latin's as Liber Regius (Al-kitab al-Malik.) The book deals with both the theory and practice of medicine. It begins with a criticism of previous Greek and Arabic medical treatises. This book was dedicated to the reigning Buwayhid ruler, Azad-up-Dawlah. It was twice translated into Latin but at last superseded by the Qanun of Ibn Sina.

Ibn Sina's reputation as a physician, Ibn Sina's Canon of Medicine was a Medical Bible:

Abu Ali Husayn ibn Sina, more commonly known to the West as Avicenna, was the greatest intellectual giant of his age whose immortal works on Medicine and Philosophy have always remained a never-failing source of guidance and inspiration not to the students of those subjects only, but to the wider reading public as well. He achieved his greatest fame in the West as a renowned physician. While still in his teens, the young medical student earned such a higher reputation that he was summoned to treat the Samanid sultan of Bukhara, Nuh-ibn-Mansur. The latter being pleased with his treatment owed to him a library. The young scholar rapidly absorbed the immense contents of the royal library and embarked upon a career of writing the age of twenty-one. He wrote a good number of books on medicine, philology, philosophy, theology, geometry, astronomy and arts. Nearly all his works are written in Arabic except a few verses. He has composed one or two treatises in Persian. His famous work on medicine known as Canon of Medicine was the most influential medical compendium to reach Europe from the Arab World. The canon was mammoth undertaking, a careful classification and systematization of all the medical knowledge known to the Arabs in the eleventh century. This medical encyclopedia deals with general medicine, diseases affecting all parts of the body, special pathology and pharmacopoeia. As regards the importance of Ibn Sina's Canon, Prof Hitti says, "The Arabic text of the Canon was published in Rome in 1593 and was therefore, one of the earliest books to see print. "Translated into Latin by the Gerard of Cremona in the 12th century, this Canon, with its encyclopedic contents, its systematic arrangement and philosophic plan, soon worked its way into a position of preeminence in the medical literature of the age displacing the works of Galen, al Razi and al-Majusi and becoming the text book of the medical education in the schools of Europe. In the last thirty years of the century, it passed through fifteen Latin editions and one Hebrew. In recent

years, a partial translation into English was made. The book distinguishes mediastinitis from pleurisy and recognizes the contagious nature and the spreading of diseases by water and soil. It gives a materia medica considers some seven hundred and sixty drugs. From the 12th to the 17th centuries, this work served as the chief guide to medical science in the west and it is still in occasional use in the Muslim East. In the words of Dr. Osler, "it has remained a medical Bible for a longer period than any other work."

Advancement of medical science:

High tribute has been paid to Ibn Sina by the Italian poet, Dante who placed him in the illustrious company of Hippocrates, Galen and Socrates. As a psychologist, Ibn Sina foreshadowed twentieth century theory on brain localization. He taught that the external senses-sight, hearing, tongue, taste and smell were centred in the brain. The Arabs emphasized upon the brain and not the heart, as the seat of reason and sense, represented an immense step forward in medical science. With Ibn Sina, Islamic medicine reached its zenith in the East.

Astronomy, Observatory erected by Mamun, Astrolabe:

The next contribution of the Arabs was in astronomy and this astronomical study was begun under the influence of an Indian work, called Siddhanta. This work was translated by Muhammad ibn Ibrahim al-Fazari at the order of Mamun. During the first half of the 9th century, the first regular observations were made at Jundhi Shapur (S.W. Persia) Mamun erected at Baghdad an astronomical observatory under the direction of a converted Jew. Sind ibn 'Ali, al-Abbas Fer-Ghani, three sons of Musa ibn Shakir's Sons made special study of astronomy. They "ascertained the size of the earth, the obliquity of the ecliptic, the variations in the lunar latitudes, the precession of equinoxes, etc." Among the astronomer-mathematicians was the famous al-Khwarizmi whose Kitab Surat al-Ard' was written in explanation of maps in the first of the ninth century. Ishaq al-Fazari constructed astrolabe.

Mathematics:

They made original contribution in the realm of mathematics. The science of Algebra, invention of decimal Arithmetic, discovery of plane and spherical geometry, the Arabic numerals and the use of zero are some of their inventions and discoveries.

Al-Khwarizmi was a great mathematician:

There were good number of mathematicians among the Muslims who shone like radiant stars in the horizon of intellectual firmament of Islam during the Middle Ages. Of the mathematicians, Muhammad ibn Musa Khwarizmi occupies an outstanding place in the history of great thinkers, who had enriched the diverse branches of knowledge during the era of early Islam. Being one of the greatest scientist of all time and the greatest of his age, Khwarizmi has made lasting contribution to the domain of mathematics, astronomy and geography. As a mathematician, he has left in effaceable marks on the pages of mathematical history of the world. He was undoubtedly one of the greatest and most original mathematicians the world has ever produced. About his celebrated works on algebra in Latin by Gerard of Cremona, the work of al-Khwarizmi was used until the 16

century as the principal mathematical text-book of European Universities and served to introduce into Europe the science of algebra. Al Khwarizmi's mathematical works were the principal source of knowledge on the subject to world for a considerable time, George Sarton pays him a high tribute when he considers him as "one of the greatest scientist of his race and the greatest of his time." He systemized the Greek and Hindu mathematical knowledge. The oldest Arithmetic composed in Arabic was known as Kitab al-Jama al-Tariq which is not available now. Al Khwarizmi was the first exponent of the use of numerals, including zero, in preference to letters. It was through him that Europe learnt the use of zero or cipher. His work on the Indian method of calculations was translated into Latin by Adelard of Bath in the 12th century. Al Khwarizmi has the distinction of being one of the founders of algebra and developed this branch of science to an exceptionally high degree. His greatest book, Hisab al-Jabr al-Muqabalah contains calculation of integration and equations presented through over 800 examples. He also introduced negative sign which were unknown to the Arabs. The translation of Khwarizmi's algebra by Chester marks an epoch of the introduction and advancement of this branch of science into Europe. "The importance of Rebert's Latin translation of Khwarizmi's algebra," says a modern Orientalist "can hardly exaggerate, because it marked the beginning of European algebra".

Al Mahani and others

(850 A.D.) was the first writer on trigonometry, Ytikub ibn Ishaq alKindi (874 A.D.) was the distinguished scholar of mathematics, astronomy and natural philosophy. He is said to have written more than two hundred books on astronomy, optics, Euclid and meteorology. Muhammad bin Isa al Mahani (884 A.D.) is credited with the invention of modern algebra. He wrote on trigonometry, astronomy, solid geometry and cubic equations. He demonstrated methods of using algebra to solve a whole field of intermediate science problems. He made observations on the solar and lunar eclipse and planetary conjunctions.

Al-Beruni:

Abu Raihan Muhammad al-Beruni (973-1049 A.D) was one of the greatest scientist of all times. He was the most original and profound thinker that Islam has ever produced in the domain of the physical and mathematical sciences. He was not only a scientist but also a historian, philosopher, naturalist, geologist, astronomer and mathematician. He had a keen geographical sense and his conclusion in that connection deserves high merit. He has written on various subjects after close investigation. Of his 15ks (1) Chronology of ancient nations, (2) History of India (Kitab al-Hind), (3) An Astronomical Encyclopedia entitled, 'Masudi Canon' and (4) A Summary of Mathematics. Astronomy and Astrology. Born in one of the suburbs of Khwarizm (Khiva), he was captured by Sultan Mahmud of Ghazni on the fall of Khiva. His Kitab al-Hind may be regarded as one of the most significant productions in the field of regional geography. He wrote his monument work, Qanun al Masudi under the patronage of Sultan Masud. On the Mathematical and astronomical roundity of the earth, the determination of its movement and gives the latitudes and longitudes of numerous places.

Umar al Khayyam, Nasiruddin Tusi:

Among the later mathematicians who were influenced by al Khwarizmi was Umar ibn Ibrahim al-Khyyam (1123 A.D.) the greatest mathematician in the 11th century. He was not only a distinguished mathematician and astronomer but also a famous poet. He had written several treatises on arithmetic, algebra and astronomy. His solution of the cubic and quadratic equations with the help of conic section is the most advanced work of Arabic mathematics. Abul Waheed Ibn Rushd (1198 A.D.), the greatest Arab Philosopher of Spain, wrote on eternity of universe, planets and eclipses. He is said to have worked on spherical trigonometry and on the theory of aeronautics also Muhammad Nasiruddin Tusi (1247. A.D.) an all round scholar with a synthetic brain, was another famous astronomer and mathematician. He worked on arithmetic, geometry, plane and spherical trigonometry and on astronomy. He gave proof of Pythagoras theorem, made observation at Maraghab in which he set up huge rings for elliptical, solstitial, armillary purposes. He edited most of the mathematical works of antiquity to the number of sixteen which practically constituted the whole scientific knowledge of the period.

Chemistry, Modern chemistry is the invention of the Muslims:

After materia medica, astronomy and mathematics, the Muslims made their greatest scientific contribution in Chemistry. Chemistry grew out of alchemy which the Islamic scientist did much to improve and advance. Their greatest contribution to world's story of knowledge was the accumulation of scientific facts and the advancement of scientific theories and methodology. "Modern Chemistry," says Mr. Humboldt, "was admittedly the inventions of Muslim, whose achievements in this sphere were of unique interest." They conclusively proved the worthlessness of ancient chemistry. They found the chemical affinities of mercury, lead, copper, silver and gold and knew the chemical process of oxidation and calcination. The Muslims were the first to teach the world distillation, filtration and crystallization. They knew how to change a liquid into vapour. It was in Muslim Spain that chemistry was first established and had the Muslims not been defeated at Poitiers, it would have reached its zenith there.

Jabir ibn Hayyan is the father of chemistry:

Jabir ibn Hayyan, known as Geber in the western world, is the father of modern chemistry. His name is the greatest in the field of medieval chemical science. He clearly recognized and stated the importance of experimentation more than any other early alchemist and made a noteworthy advance in theory and practice of chemistry. He wrote some five hundred treatises on chemistry. His works after the 14th century scientifically the two principal operations of chemistry, calcination and reduction. Jabir improved on the methods for evaporation, sublimation, melting and crystallization. He discovered for the first time nitric acid, sulphuric acid, aqua regia, silver nitrate and several other compounds and new how to produce aqua regia in which gold and silver could be dissolved. He was the prophet and forerunner of positive and dynamic science, from his studies, he was able to predict the vast part which gases would be found to play to the make-up and composition of elements and more complex substances. He brought

system and order to chemical science and made it almost a sacred art. He was followed by others whose originality and industry, profoundness of knowledge and keenness of observation evoke astonishment of the readers.

Al-Razi & Ibn Sina:

Other important chemist of that era were al-Razi (Rhazes) who re-discovered sulphuric acid and aqua-vitae and Ibn Sina (Avicenna) a renowned doctor, who helped to systematize medical chemistry. Later on, al-Razi became celebrated all over Europe and was considered the foremost authority and master of chemical science by all men of learning.

In spite of overwhelming evidence to the contrary, European historians have deliberately exaggerated the so-called inferiority of Arabian science. Ignorance and superstition reigned supreme in Europe while the Arab peoples were attaining a literary and scientific renaissance. When the Arabs were making epoch-making discoveries in their world, the Europeans were placing a premium upon every conceivable kind of social and political evil. A case in point is the field of chemistry which was considered one of the most significant realms of science of the Arabs. They called it "the science of key". As the science of chemistry advanced, the other sciences progressed apace.

The Muslims savants not only kept themselves confined to the advancement of science but also they made great contributions in on other braches of knowledge.

Philosophy:

The Muslims cultivated philosophy with as much zeal as the exact sciences. Al-Ghazali, al-Kindi al-Farabi was called by the Arabs a second Aristotle. He wrote various psychological political and metaphysical, works. Ibn Sina was a philosopher, physician, philologist and poet. He systematized the philosophies of his predecessors, both Muslims and Greeks. -He was the forerunner of compendium writers for the whole world" with him ended the great age Arab philosophy.

Geography:

The Arab Muslims invented the "mariners' compass" and voyaged to all parts of the world in search of knowledge or in pursuit of commerce. They even penetrated as far as America". Voyage and pilgrimage gave them inspiration to the study of geography, Ibn Khurdebeh Jaihani, al-Masudi AlIstaqri, Ibn-Hawkal, Yakut, al-Bakri, al-Muqaddsi and Idrisi are the most famous of Arabs geographers.

Literature, Jurisprudence:

The eminent scholars who shed lustre on Arabic and Persian literatures are Ispaham, ibn-Khalikan, Abu Nuwas, Al-Buhtari, Mutrannabi, Dakiki, Firdausi, Unsaury Jalaluddin, and Abul Firaj Muhammad Bin Isaq. It was as this period that the Muslims "after the Roman cultivated the science of jurisprudence and evolved there from an independent system". Their system, fiqh, was first based on the Quran and the Hadith. But when it could not solved the complicated problems, private interpretation became essential. The permission of private interpretation culminated in the establishment of four orthodox schools headed